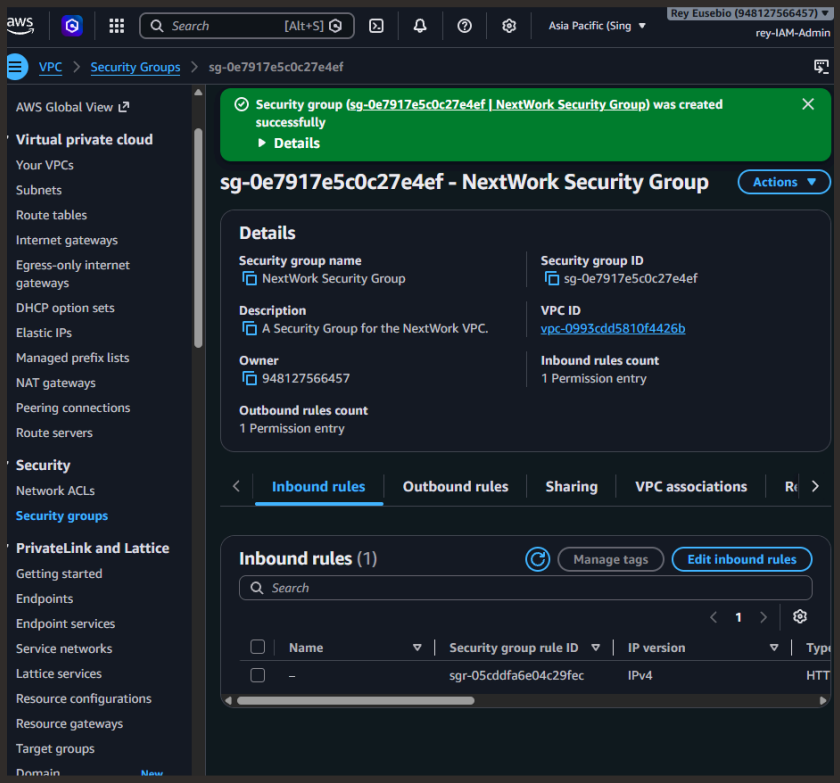
VPC TRAFFIC FLOW AND SECURITY

By Reynaldo Eusebio | September 2026



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual network in AWS, and it is useful because it allows you to create and manage your own network environment, including subnets, route tables, and internet gateways.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create and manage a virtual network with subnets, an Internet Gateway, route table, security group, and Network ACL.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how many different networking resources, such as route tables, security groups, and Network ACLs, work together to control traffic in a VPC.

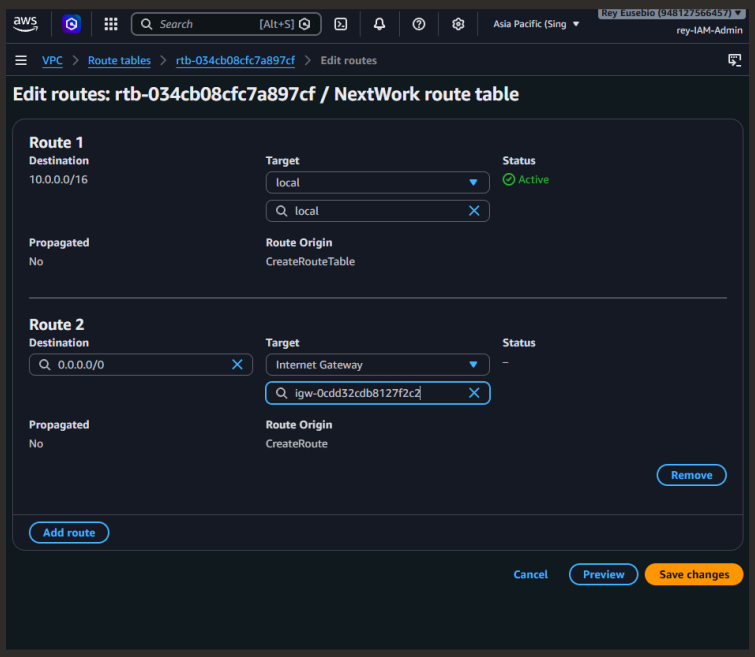
This project took me...

This project took me approximately 3 hours because I spent time understanding and setting up the VPC, route table, security group, Network ACL, EC2 Global View, and resources in another AWS Region.

Route tables

Route tables are used to route traffic by determining where network traffic should go.

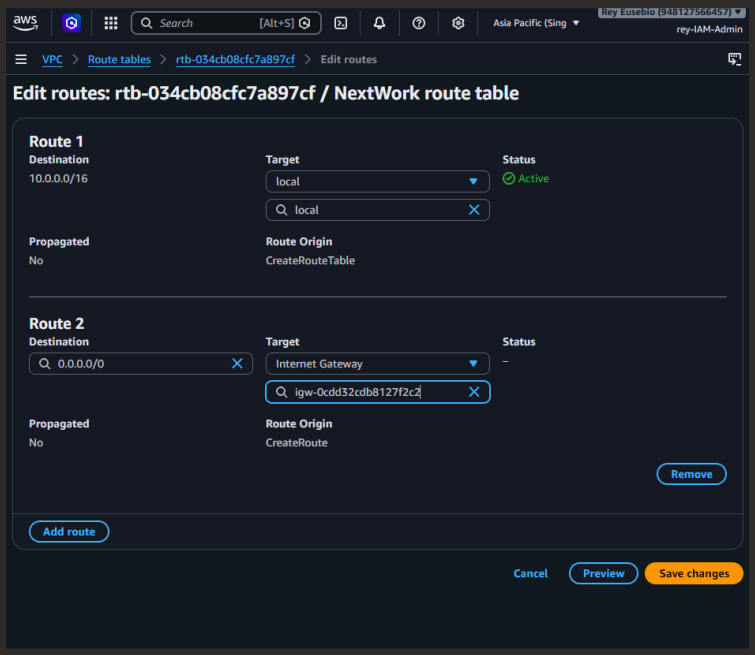
Route tables are needed to make a subnet public because they can direct traffic to the Internet Gateway.



Route destination and target

Routes are defined by their destination and target, which means they guide traffic where it should go.

The route in my route table that directed internet-bound traffic to my Internet Gateway had a destination of any IP address and a target of the Internet Gateway.



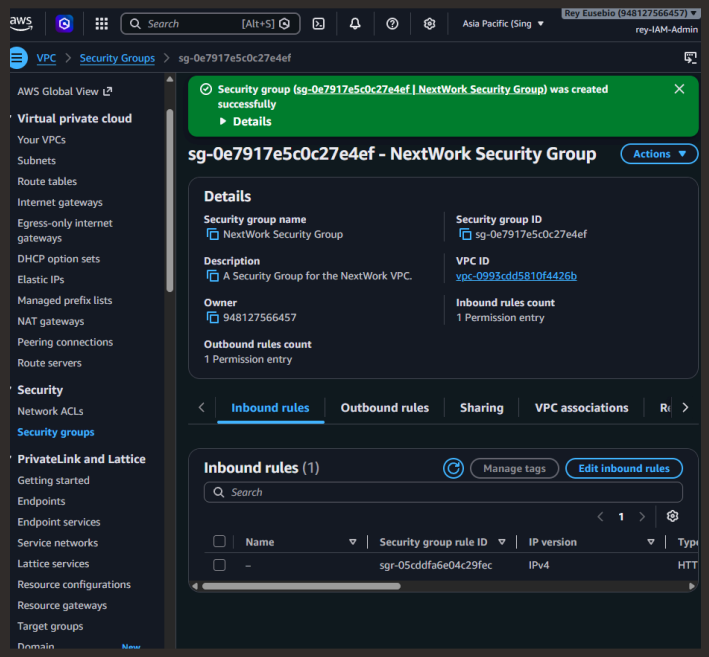
Security groups

Security groups are virtual firewalls that control inbound and outbound traffic to AWS resources.

Inbound vs Outbound rules

Inbound rules are rules that control traffic entering a resource. I configured an inbound rule that allows HTTP traffic from anywhere using port 80.

Outbound rules are rules that control traffic leaving a resource. By default, my security group's outbound rule allows all traffic to leave the resource.



Network ACLs

Network ACLs are used to guard the subnet.

Security groups vs. network ACLs

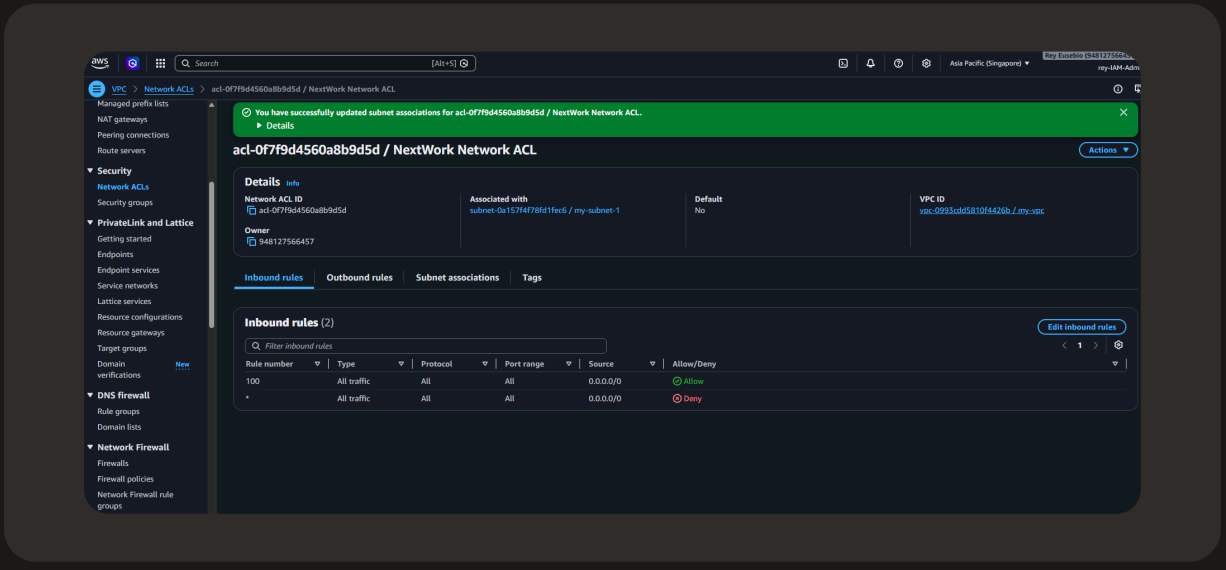
The difference between a security group and a network ACL is that a security group provides security at the resource level, while a network ACL provides security at the subnet level.

Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules allow all traffic.

In contrast, a custom ACL's inbound and outbound rules are automatically set to deny.



Tracking VPC Resources

I created an additional VPC, Internet Gateway, and security group in another Region, Ohio, instead of my usual Region. Teams can use multiple Regions to serve users from different locations.

EC2 Global View is a tool where you can find VPC resources across all AWS Regions. I could even narrow down my search by viewing the resources in a specific Region. Without EC2 Global View, you'd have to switch between Regions and check your resources individually.

Now that I've learnt about EC2 Global View, I'd use it again to view and manage my EC2 and VPC resources across all AWS Regions from a single dashboard.

